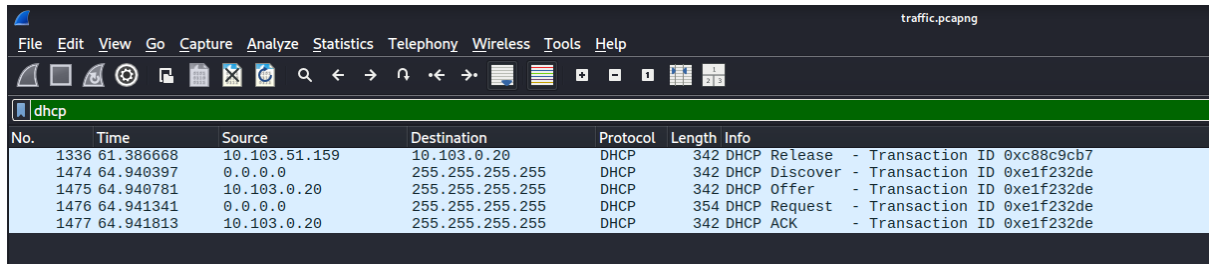


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Course-Ethical Hacking(JOVAC-2026)

Wireshark Assignment

Q1. Analyze the provided network capture file and determine the total number of **DHCP** messages.

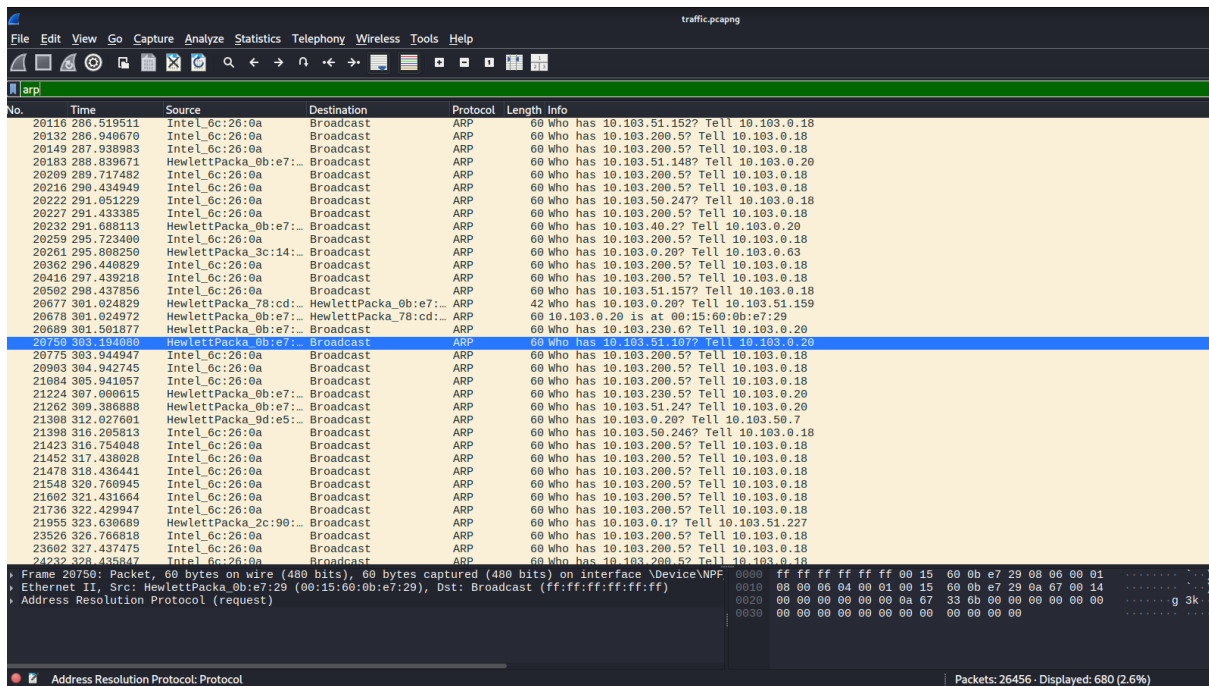
Answer: 5



No.	Time	Source	Destination	Protocol	Length	Info
1336	61.386668	10.103.51.159	10.103.0.20	DHCP	342	DHCP Release - Transaction ID 0xc88c9cb7
1474	64.940397	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xe1f232de
1475	64.940781	10.103.0.20	255.255.255.255	DHCP	342	DHCP Offer - Transaction ID 0xe1f232de
1476	64.941341	0.0.0.0	255.255.255.255	DHCP	354	DHCP Request - Transaction ID 0xe1f232de
1477	64.941813	10.103.0.20	255.255.255.255	DHCP	342	DHCP ACK - Transaction ID 0xe1f232de

Q2. Analyze the network capture file and determine the total number of **ARP** messages.

Answer: 680



No.	Time	Source	Destination	Protocol	Length	Info
20116	286.519511	Intel_6c:26:0a	Broadcast	ARP	60	Who has 10.103.51.152? Tell 10.103.0.18
20132	286.940670	Intel_6c:26:0a	Broadcast	ARP	60	Who has 10.103.200.57 Tell 10.103.0.18
20149	287.938983	Intel_6c:26:0a	Broadcast	ARP	60	Who has 10.103.200.57 Tell 10.103.0.18
20183	288.830671	HewlettPacka_0b:e7:...	Broadcast	ARP	60	Who has 10.103.51.148? Tell 10.103.0.20
20209	289.717482	Intel_6c:26:0a	Broadcast	ARP	60	Who has 10.103.200.57 Tell 10.103.0.18
20216	290.434949	Intel_6c:26:0a	Broadcast	ARP	60	Who has 10.103.200.57 Tell 10.103.0.18
20222	291.051229	Intel_6c:26:0a	Broadcast	ARP	60	Who has 10.103.50.247? Tell 10.103.0.18
20227	291.433385	Intel_6c:26:0a	Broadcast	ARP	60	Who has 10.103.200.57 Tell 10.103.0.18
20232	291.688113	HewlettPacka_0b:e7:...	Broadcast	ARP	60	Who has 10.103.40.27 Tell 10.103.0.20
20259	295.723400	Intel_6c:26:0a	Broadcast	ARP	60	Who has 10.103.200.57 Tell 10.103.0.18
20261	295.808250	HewlettPacka_3c:14:...	Broadcast	ARP	60	Who has 10.103.0.20? Tell 10.103.0.63
20362	296.440829	Intel_6c:26:0a	Broadcast	ARP	60	Who has 10.103.200.57 Tell 10.103.0.18
20416	297.439218	Intel_6c:26:0a	Broadcast	ARP	60	Who has 10.103.200.57 Tell 10.103.0.18
20502	298.437856	Intel_6c:26:0a	Broadcast	ARP	60	Who has 10.103.51.157? Tell 10.103.0.18
20677	301.024829	HewlettPacka_78:cd:...	HewlettPacka_0b:e7:...	ARP	42	Who has 10.103.0.20? Tell 10.103.51.159
20678	301.024972	HewlettPacka_0b:e7:...	HewlettPacka_78:cd:...	ARP	60	10.103.0.20 is at 00:15:00:0b:e7:29
20689	301.501877	HewlettPacka_0b:e7:...	Broadcast	ARP	60	Who has 10.103.230.67 Tell 10.103.0.20
20750	303.194800	HewlettPacka_0b:e7:...	Broadcast	ARP	60	Who has 10.103.51.107? Tell 10.103.0.20
20775	303.944947	Intel_6c:26:0a	Broadcast	ARP	60	Who has 10.103.200.57 Tell 10.103.0.18
20903	304.942745	Intel_6c:26:0a	Broadcast	ARP	60	Who has 10.103.200.57 Tell 10.103.0.18
21084	305.941057	Intel_6c:26:0a	Broadcast	ARP	60	Who has 10.103.200.57 Tell 10.103.0.18
21224	307.000615	HewlettPacka_0b:e7:...	Broadcast	ARP	60	Who has 10.103.230.57 Tell 10.103.0.20
21262	309.386888	HewlettPacka_0b:e7:...	Broadcast	ARP	60	Who has 10.103.51.247 Tell 10.103.0.20
21308	312.927091	HewlettPacka_9d:e5:...	Broadcast	ARP	60	Who has 10.103.0.20? Tell 10.103.90.7
21390	316.205813	Intel_6c:26:0a	Broadcast	ARP	60	Who has 10.103.50.246? Tell 10.103.0.18
21423	316.754048	Intel_6c:26:0a	Broadcast	ARP	60	Who has 10.103.200.57 Tell 10.103.0.18
21452	317.438028	Intel_6c:26:0a	Broadcast	ARP	60	Who has 10.103.200.57 Tell 10.103.0.18
21478	318.436441	Intel_6c:26:0a	Broadcast	ARP	60	Who has 10.103.200.57 Tell 10.103.0.18
21548	320.760945	Intel_6c:26:0a	Broadcast	ARP	60	Who has 10.103.200.57 Tell 10.103.0.18
21602	321.431064	Intel_6c:26:0a	Broadcast	ARP	60	Who has 10.103.200.57 Tell 10.103.0.18
21736	322.429947	Intel_6c:26:0a	Broadcast	ARP	60	Who has 10.103.200.57 Tell 10.103.0.18
21955	323.630689	HewlettPacka_2c:90:...	Broadcast	ARP	60	Who has 10.103.0.17 Tell 10.103.51.227
23526	326.766818	Intel_6c:26:0a	Broadcast	ARP	60	Who has 10.103.200.57 Tell 10.103.0.18
23602	327.437475	Intel_6c:26:0a	Broadcast	ARP	60	Who has 10.103.200.57 Tell 10.103.0.18
24232	328.438047	Intel_6c:26:0a	Broadcast	ARP	60	Who has 10.103.200.57 Tell 10.103.0.18

Frame 20750: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface Device\NPF...
Ethernet II, Src: HewlettPacka_0b:e7:29 (00:15:00:0b:e7:29), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Address Resolution Protocol (request)

Address Resolution Protocol: Protocol

Packets: 26456 - Displayed: 680 (2.6%)

Q3. Analyze the network capture file and identify the **IP address** that accessed **baidu.com**.

Answer: 10.103.51.159

```

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
[Icons]
[Filter] [Hosts] [Packets] [Bytes] [Time] [Length] [Info]
[Hosts] [Packets] [Bytes] [Time] [Length] [Info]
No. Time Source Destination Protocol Length Info
176 10.495344 10.103.101.3 224.0.0.251 NDNS 113 Standard query 0x0900 PTR _communicator._tcp.local, "QM" question OPT
177 10.495499 fe80::da38:fcff::fe2 ffb2::fb NDNS 133 Standard query 0x0900 PTR _communicator._tcp.local, "QM" question OPT
180 10.708903 10.103.50.171 239.255.255.250 SSDP 216 M-SEARCH * HTTP/1.1
181 10.804232 10.103.230.1 239.255.255.250 NDNS 92 Name query NB VUE<ic>
183 10.990765 10.103.50.34 10.103.255.255 SMB_NE_ 282 SAN LOGON request from client[Malformed Packet]
184 10.103570 10.103.50.171 10.103.255.255 NDNS 92 Name query NB SETH-02<ic>
187 11.223953 10.103.50.172 10.103.255.255 NDNS 92 Name query NB BDOMAIN<ic>
188 11.224879 10.103.50.172 10.103.255.255 SMB_NE_ 282 SAN LOGON request from client[Malformed Packet]
189 11.243852 10.103.230.13 10.103.255.255 NDNS 92 Name query NB BDOMAIN<ic>
191 11.301384 1d:00:00:0e:00:00:00:00 902.11 376 Power-Save poll, Flags: pmP, FT
193 11.320047 10.103.50.3 239.255.255.250 SSDP 216 M-SEARCH * HTTP/1.1
194 11.564955 10.103.230.1 10.103.255.255 NDNS 92 Name query NB VUE<ic>
196 11.723063 10.103.50.171 239.255.255.250 SSDP 216 M-SEARCH * HTTP/1.1
196 11.732475 10.103.50.133 239.255.255.250 SSDP 216 M-SEARCH * HTTP/1.1
198 12.619927 10.103.230.13 10.103.255.255 NDNS 92 Name query NB VUE<ic>
198 12.619312 10.103.50.32 10.103.255.255 NDNS 92 Name query NB BDOMAIN<ic>
201 12.182652 10.103.50.32 10.103.255.255 SMB_NE_ 282 SAN LOGON request from client[Malformed Packet]
203 12.303753 Xerox-00:10:10:00 0x003a 802.11 377 PVA Management[Malformed Packet]
204 12.458991 10.103.230.1 10.103.255.255 NDNS 92 Name query NB VUE<ic>
224 12.723632 10.103.50.171 239.255.255.250 SSDP 216 M-SEARCH * HTTP/1.1
225 12.736634 10.103.50.133 239.255.255.250 SSDP 216 M-SEARCH * HTTP/1.1
226 12.762429 10.103.230.13 10.103.255.255 NDNS 92 Name query NB VUE<ic>
229 13.409871 10.103.101.2 224.0.0.251 NDNS 84 Standard query 0x0900 PTR _communicator._tcp.local, "QM" question
230 13.410171 fe80::2258:69ff::fe2 ffb2::fb NDNS 104 Standard query 0x0900 PTR _communicator._tcp.local, "QM" question
232 13.738575 10.103.50.171 239.255.255.250 SSDP 216 M-SEARCH * HTTP/1.1
233 13.751268 10.103.50.171 239.255.255.250 SSDP 216 M-SEARCH * HTTP/1.1
235 13.945999 10.103.50.26 10.103.255.255 NDNS 92 Name query NB SETH-02<ic>
254 14.009218 10.103.230.5 10.103.255.255 NDNS 92 Name query NB VUE<ic>
255 14.336022 10.103.101.2 224.0.0.251 NDNS 113 Standard query 0x0900 PTR _communicator._tcp.local, "QM" question
257 14.421008 fe80::2258:69ff::fe2 ffb2::fb NDNS 104 Standard query 0x0900 PTR _communicator._tcp.local, "QM" question
258 14.450616 10.103.101.3 10.103.255.255 802.11 377 Unknown protocol version: 2[Malformed Packet]
259 14.450819 10.103.4.9 255.255.255.255 TFTF 63 Read Request, File: class409.cfg, Transfer type: octet
260 14.483400 10.103.101.2 10.103.255.255 NDNS 282 SAN LOGON request from client[Malformed Packet]
261 14.535077 10.103.50.181 10.103.255.255 SMB_NE_ 280 SAN LOGON request from client[Malformed Packet]
261 14.535077 10.103.50.181 10.103.255.255 SMB_NE_ 280 SAN LOGON request from client[Malformed Packet]
Frame 261: 1452 bytes on wire (2256 bits) captured (2256 bits) on interface Device\NPF...
Ethernet II, Src: MicroStarNt-75:2e:a4:30:3c:23;75:2e:a4, Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Internet Protocol Version 4, Src: 10.103.50.32, Dst: 10.103.255.255
0100 .... 0 Version: 4
... .. 0 Header length: 20 bytes (5)
... .. 0 Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 208
... .. 0
... .. 0
Header length in 32-bit words (ip_hdr_len), 4 bits
Packets: 26456 - Displayed: 9102 (34.4%)

```

Answer: 578

Q7. Determine the total number of packets **sent to** the IP address **10.103.0.20**.

Answer:547

Q8. Apply a filter to display only **IPv4** traffic and determine the total number of packets shown.

Answer: 24906

ip.version=4						
No.	Time	Source	Destination	Protocol	Length Info	
20615	300.677927	142.250.186.78	10.103.51.159	QUIC	1392 Protected Payload (KPo)	
20616	300.677991	10.103.51.159	142.250.186.78	QUIC	77 Protected Payload (KPo)	
20617	300.678044	142.250.186.78	10.103.51.159	QUIC	1392 Protected Payload (KPo)	
20618	300.678161	142.250.186.78	10.103.51.159	QUIC	1392 Protected Payload (KPo)	
20619	300.678177	10.103.51.159	142.250.186.78	QUIC	75 Protected Payload (KPo)	
20620	300.678262	142.250.186.78	10.103.51.159	QUIC	1392 Protected Payload (KPo)	
20621	300.678306	10.103.51.159	142.250.186.78	QUIC	75 Protected Payload (KPo)	
20622	300.678378	142.250.186.78	10.103.51.159	QUIC	1392 Protected Payload (KPo)	
20623	300.679395	142.250.186.78	10.103.51.159	QUIC	1392 Protected Payload (KPo)	
20624	300.679418	10.103.51.159	142.250.186.78	QUIC	75 Protected Payload (KPo)	
20625	300.679418	142.250.186.78	10.103.51.159	QUIC	1392 Protected Payload (KPo)	
20626	300.679506	142.250.186.78	10.103.51.159	QUIC	1392 Protected Payload (KPo)	
20627	300.679565	10.103.51.159	142.250.186.78	QUIC	75 Protected Payload (KPo)	
20628	300.680848	142.250.186.78	10.103.51.159	QUIC	1392 Protected Payload (KPo)	
20629	300.680998	142.250.186.78	10.103.51.159	QUIC	1392 Protected Payload (KPo)	
20630	300.681121	10.103.51.159	142.250.186.78	QUIC	75 Protected Payload (KPo)	
20631	300.682534	142.250.186.78	10.103.51.159	QUIC	392 Protected Payload (KPo)	
20632	300.687499	10.103.51.159	10.103.0.29	DNS	75 Standard query 0x1835 A ssl.gstatic.com	
20633	300.687754	10.103.0.29	10.103.51.159	DNS	91 Standard query response 0x1835 A ssl.gstatic.com A 142.250.186.67	
20634	300.689473	10.103.51.159	142.250.186.67	QUIC	1392 Initial, DCID=5bd88ac01024cf31, PKN: 1, CRYPTO, PADDING	
20635	300.709113	10.103.51.159	142.250.186.78	QUIC	75 Protected Payload (KPo)	
20636	300.757450	142.250.186.67	10.103.51.159	QUIC	1392 Protected Payload (KPo)	
20637	300.758152	10.103.51.159	142.250.186.67	QUIC	206 Protected Payload (KPo)	
20638	300.758532	142.250.186.78	10.103.51.159	QUIC	68 Protected Payload (KPo)	
20639	300.773609	10.103.51.159	142.250.184.228	QUIC	315 Protected Payload (KPo)	
20640	300.815862	10.103.51.159	142.250.185.110	QUIC	1392 Initial, DCID=1c386a935bdac3cc, PKN: 1, CRYPTO, PADDING	
20641	300.817591	142.250.186.67	10.103.51.159	QUIC	664 Protected Payload (KPo)	
20642	300.817591	142.250.186.67	10.103.51.159	QUIC	119 Protected Payload (KPo)	
20643	300.817862	10.103.51.159	142.250.186.67	QUIC	75 Protected Payload (KPo)	
20644	300.829112	142.250.184.228	10.103.51.159	QUIC	71 Protected Payload (KPo)	
20645	300.842882	142.250.186.67	10.103.51.159	QUIC	67 Protected Payload (KPo)	
20646	300.844963	10.103.51.159	142.250.184.228	QUIC	75 Protected Payload (KPo)	
20647	300.873811	10.103.51.101	239.255.255.250	SSDP	210 M-SEARCH * HTTP/1.1	
20648	300.879874	142.250.185.110	10.103.51.159	QUIC	1392 Protected Payload (KPo)	
20649	300.880473	10.103.51.159	142.250.185.110	QUIC	287 Protected Payload (KPo)	

Q9. Examine the last SMB packet in the capture and identify its source IP address.

Answer: 10.103.50.202

smb						
No.	Time	Source	Destination	Protocol	Length Info	
21487	320.346955	10.103.50.216	10.103.255.255	SMB_NE...	280 SAM LOGON request from client[Malformed Packet]	
21512	320.234853	10.103.50.14	10.103.255.255	BROWSER	243 Host Announcement 409_06, Workstation, Server, SQL Server, NT Workstation	
21533	320.506901	10.103.50.116	10.103.255.255	SMB_NE...	280 SAM LOGON request from client[Malformed Packet]	
21739	322.464767	10.103.50.164	10.103.255.255	SMB_NE...	280 SAM LOGON request from client[Malformed Packet]	
22851	324.461648	10.103.51.147	10.103.255.255	SMB_NE...	280 SAM LOGON request from client[Malformed Packet]	
23499	326.108830	10.103.51.149	10.103.255.255	SMB_NE...	280 SAM LOGON request from client[Malformed Packet]	
23550	327.026571	10.103.50.81	10.103.255.255	SMB_NE...	280 SAM LOGON request from client[Malformed Packet]	
25160	329.283275	10.103.51.150	10.103.255.255	BROWSER	243 Host Announcement 409_13, Workstation, Server, SQL Server, NT Workstation	
25198	329.505032	10.103.50.214	10.103.255.255	SMB_NE...	280 SAM LOGON request from client[Malformed Packet]	
25201	329.551542	10.103.51.96	10.103.255.255	SMB_NE...	280 SAM LOGON request from client[Malformed Packet]	
25217	330.209007	10.103.51.144	10.103.255.255	SMB_NE...	280 SAM LOGON request from client[Malformed Packet]	
25229	331.062627	10.103.51.29	10.103.255.255	BROWSER	243 Host Announcement 409_12, Workstation, Server, SQL Server, NT Workstation	
25259	331.425749	10.103.50.173	10.103.255.255	SMB_NE...	280 SAM LOGON request from client[Malformed Packet]	
25261	331.488641	10.103.51.101	10.103.255.255	SMB_NE...	280 SAM LOGON request from client[Malformed Packet]	
25266	331.639899	10.103.50.72	10.103.255.255	SMB_NE...	280 SAM LOGON request from client[Malformed Packet]	
25353	336.069558	10.103.51.132	10.103.255.255	SMB_NE...	282 SAM LOGON request from client[Malformed Packet]	
26070	337.820621	10.103.51.152	10.103.255.255	BROWSER	243 Host Announcement 409_18, Workstation, Server, SQL Server, NT Workstation	
26206	338.406746	10.103.51.151	10.103.255.255	BROWSER	243 Host Announcement 409_19, Workstation, Server, SQL Server, NT Workstation	
26285	339.125395	10.103.51.154	10.103.255.255	SMB_NE...	280 SAM LOGON request from client[Malformed Packet]	
26291	339.277822	10.103.50.40	10.103.255.255	SMB_NE...	280 SAM LOGON request from client[Malformed Packet]	
26313	339.883735	10.103.50.175	10.103.255.255	BROWSER	243 Host Announcement SETB-03, Workstation, Server, NT Workstation	
26338	341.144851	10.103.50.188	10.103.255.255	SMB_NE...	280 SAM LOGON request from client[Malformed Packet]	
26341	341.441479	10.103.50.183	10.103.255.255	SMB_NE...	280 SAM LOGON request from client[Malformed Packet]	
26343	342.201530	10.103.51.92	10.103.255.255	SMB_NE...	280 SAM LOGON request from client[Malformed Packet]	
26351	342.790849	10.103.50.213	10.103.255.255	SMB_NE...	280 SAM LOGON request from client[Malformed Packet]	
26353	342.881173	10.103.50.166	10.103.255.255	SMB_NE...	280 SAM LOGON request from client[Malformed Packet]	
26364	344.203370	10.103.50.43	10.103.255.255	SMB_NE...	280 SAM LOGON request from client[Malformed Packet]	
26379	344.869709	10.103.50.165	10.103.255.255	SMB_NE...	280 SAM LOGON request from client[Malformed Packet]	
26431	345.760679	10.103.51.97	10.103.255.255	SMB_NE...	280 SAM LOGON request from client[Malformed Packet]	
26440	346.495923	10.103.50.200	10.103.255.255	SMB_NE...	280 SAM LOGON request from client[Malformed Packet]	
26441	346.497230	10.103.51.154	10.103.255.255	BROWSER	243 Host Announcement 409_16, Workstation, Server, SQL Server, NT Workstation	
26444	346.615864	10.103.50.248	10.103.255.255	BROWSER	243 Host Announcement SETB-11, Workstation, Server, NT Workstation	
26451	347.483516	10.103.50.84	10.103.255.255	SMB_NE...	280 SAM LOGON request from client[Malformed Packet]	
26452	347.579883	10.103.50.202	10.103.255.255	SMB_NE...	280 SAM LOGON request from client[Malformed Packet]	

Q10. Identify the MAC address associated with the IP address 151.139.128.14.

Answer: 00:1c:7f:6c:96:3f

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

ip.addr == 151.139.128.14

No.	Time	Source	Destination	Protocol	Length	Info
19004	228.211534	10.103.51.159	151.139.128.14	TCP	66	64079 → 80 [SYN] Seq=0 Win=64
19005	228.265841	151.139.128.14	10.103.51.159	TCP	66	80 → 64079 [SYN, ACK] Seq=0 A
19006	228.265945	10.103.51.159	151.139.128.14	TCP	54	64079 → 80 [ACK] Seq=1 Ack=1
19007	228.266244	10.103.51.159	151.139.128.14	HTTP	290	GET /MFEwTzBNMESwSTAJBgUrDgMC
19008	228.320802	151.139.128.14	10.103.51.159	TCP	60	80 → 64079 [ACK] Seq=1 Ack=23
19009	228.323598	151.139.128.14	10.103.51.159	TCP	499	80 → 64079 [PSH, ACK] Seq=1 A
19010	228.324235	151.139.128.14	10.103.51.159	OCSP	781	Response
19011	228.324263	10.103.51.159	151.139.128.14	TCP	54	64079 → 80 [ACK] Seq=237 Ack=
19012	228.332276	10.103.51.159	151.139.128.14	HTTP	292	GET /MFEwTzBNMESwSTAJBgUrDgMC
19015	228.386257	151.139.128.14	10.103.51.159	TCP	60	80 → 64079 [ACK] Seq=1173 Ack=
19016	228.389657	151.139.128.14	10.103.51.159	TCP	499	80 → 64079 [PSH, ACK] Seq=117
19017	228.389784	151.139.128.14	10.103.51.159	OCSP	563	Response
19018	228.389815	10.103.51.159	151.139.128.14	TCP	54	64079 → 80 [ACK] Seq=475 Ack=
20241	293.387153	151.139.128.14	10.103.51.159	TCP	60	80 → 64079 [FIN, ACK] Seq=212
20242	293.387221	10.103.51.159	151.139.128.14	TCP	54	64079 → 80 [ACK] Seq=475 Ack=
20243	293.387275	10.103.51.159	151.139.128.14	TCP	54	64079 → 80 [FIN, ACK] Seq=475
20244	293.440908	151.139.128.14	10.103.51.159	TCP	60	80 → 64079 [ACK] Seq=2128 Ack=

Frame 19004: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF...
 Ethernet II, Src: HewlettPacka_78:cd:4a (10:60:4b:78:cd:4a), Dst: CheckPointSo_6c:96:3f (00:1c:7f:6c:96:3f)
 Destination: CheckPointSo_6c:96:3f (00:1c:7f:6c:96:3f)
 = LG bit: Globally unique address (factory default)
 = IG bit: Individual address (unicast)
 Source: HewlettPacka_78:cd:4a (10:60:4b:78:cd:4a)
 = LG bit: Globally unique address (factory default)
 = IG bit: Individual address (unicast)

Specifies if this is an individual (unicast) or group (broadcast/multicast) address (eth.dst.ig), 1 bit

Q11. Examine **packet number 416** and identify the **protocol** being used.

Answer: DNS

Q12. Examine **packet number 416** and identify its **destination IP address**.

Answer:10.103.0.20

Q13. Identify the **source IP address** of the **first HTTP packet** in the network capture.

Answer:10.103.51.159

Q14. Identify the **source port** of the **first HTTP packet** in the network capture.

Answer: 64079

http						
No.	Time	Source	Destination	Protocol	Length	Info
19007	228.266244	10.103.51.159	151.139.128.14	HTTP	290	GET /MFEwTzBNMESwSTAJBgUrDg
19010	228.324235	151.139.128.14	10.103.51.159	OCSP	781	Response
19012	228.332276	10.103.51.159	151.139.128.14	HTTP	292	GET /MFEwTzBNMESwSTAJBgUrDg
19017	228.389784	151.139.128.14	10.103.51.159	OCSP	563	Response
19802	269.746721	10.103.51.159	23.221.154.40	HTTP	271	GET /singletile/summary/ali
19804	269.749357	23.221.154.40	10.103.51.159	HTTP	315	HTTP/1.1 301 Moved Permanen
20767	303.694964	10.103.51.159	103.235.46.39	HTTP	1169	GET / HTTP/1.1
20885	304.818396	10.103.51.159	103.235.46.39	HTTP	1228	GET /img/PCtm_d9c8750bed0b3
20886	304.819122	10.103.51.159	103.235.46.39	HTTP	1214	GET /img/flexible/logo/pc/r
20891	304.821164	103.235.46.39	10.103.51.159	HTTP	1409	HTTP/1.1 200 OK (text/html
20892	304.821788	10.103.51.159	103.235.46.39	HTTP	1216	GET /img/flexible/logo/pc/r
20919	305.181055	10.103.51.159	103.235.46.39	HTTP	1219	GET /img/flexible/logo/pc/p
20937	305.186294	103.235.46.39	10.103.51.159	HTTP	1491	HTTP/1.1 200 OK (PNG)
20951	305.187355	103.235.46.39	10.103.51.159	HTTP	309	HTTP/1.1 200 OK (PNG)
20960	305.190320	103.235.46.39	10.103.51.159	HTTP	1163	HTTP/1.1 200 OK (PNG)
21007	305.544293	103.235.46.39	10.103.51.159	HTTP	1053	HTTP/1.1 200 OK (PNG)
21040	305.694886	10.103.51.159	110.242.69.147	HTTP	467	GET /static/superman/img/qr
21043	305.710660	10.103.51.159	110.242.69.147	HTTP	405	GET /static/superman/js/lib
21046	305.719204	10.103.51.159	110.242.69.147	HTTP	469	GET /static/superman/img/to
21049	305.722190	10.103.51.159	110.242.69.147	HTTP	467	GET /static/superman/img/to
21052	305.731433	10.103.51.159	110.242.69.147	HTTP	466	GET /static/superman/img/to
21058	305.776124	10.103.51.159	110.242.69.147	HTTP	467	GET /static/superman/img/to
21061	305.776754	10.103.51.159	110.242.69.147	HTTP	431	GET /static/superman/font/i
21074	305.890442	10.103.51.159	123.52.189.38	HTTP	1018	GET /cd37ed75a9387c5b.js HT
21088	305.981160	110.242.69.147	10.103.51.159	HTTP	176	[TCP Previous segment not c
21095	306.018200	110.242.69.147	10.103.51.159	HTTP	1231	[TCP Previous segment not c
21097	306.018486	110.242.69.147	10.103.51.159	HTTP	1230	[TCP Out-Of-Order] Continua
21099	306.018783	110.242.69.147	10.103.51.159	HTTP	522	[TCP Previous segment not c
21102	306.025809	110.242.69.147	10.103.51.159	HTTP	584	[TCP Previous segment not c
21104	306.025972	110.242.69.147	10.103.51.159	HTTP	1514	[TCP Out-Of-Order] Continua
21109	306.027535	10.103.51.159	110.242.69.147	HTTP	473	GET /static/superman/img/to
21112	306.032059	110.242.69.147	10.103.51.159	HTTP	999	[TCP Previous segment not c
21114	306.032141	110.242.69.147	10.103.51.159	HTTP	1230	[TCP Out-Of-Order] Continua
21121	306.057340	110.242.69.147	10.103.51.159	HTTP	136	[TCP Previous segment not c
21129	306.146000	110.242.69.147	10.103.51.159	HTTP	1022	[TCP Previous segment not c
Header Checksum: 0x0000 [validation disabled]						
[Header checksum status: Unverified]						
Source Address: 10.103.51.159						
Destination Address: 151.139.128.14						
[Stream index: 388]						
Transmission Control Protocol, Src Port: 64079, Dst Port: 80, Seq: 1, Ack: 1, Len: 236						
Source Port: 64079						
Destination Port: 80						
Source Port (tcp.srcport), 2 bytes						

Q15. Determine the total **capture duration** of the network file (in **seconds**).

Answer: 348 seconds

Time	
First packet:	2021-07-26 08:53:50
Last packet:	2021-07-26 08:59:38
Elapsed:	00:05:48

Q16. Determine the total number of **NBNS** packets present in the network capture.

Answer:963

Q17. Determine the total number of **TCP** packets with **source port 443**.

Answer:7275

tcp.srcport==443						
No.	Time	Source	Destination	Protocol	Length	Info
127	9.188070	204.79.197.200	10.103.51.159	TCP	60	59023 → 443 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
130	9.188000	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=1 Ack=490 Win=4204800 Len=0
131	9.188529	204.79.197.200	10.103.51.159	TLSv1.2	204	Server Hello, Change Cipher Spec, Encrypted Handshake Message
134	9.190123	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=151 Ack=541 Win=4204544 Len=0
135	9.190275	204.79.197.200	10.103.51.159	TLSv1.2	123	Application Data
142	9.192287	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=220 Ack=628 Win=4204544 Len=0
143	9.192287	204.79.197.200	10.103.51.159	TLSv1.2	92	Application Data
144	9.192287	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=258 Ack=666 Win=4204544 Len=0
146	9.192569	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=258 Ack=2106 Win=4204800 Len=0
147	9.192618	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=258 Ack=2211 Win=4204544 Len=0
148	9.192678	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=258 Ack=2717 Win=4204032 Len=0
151	9.271024	204.79.197.200	10.103.51.159	TLSv1.2	228	Application Data
327	17.697921	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=432 Ack=2786 Win=4204032 Len=0
330	17.723248	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=432 Ack=3070 Win=4203776 Len=0

Q18. Determine the total number of TCP packets with destination port 443.

Answer:5966

tcp.dstport==443						
No.	Time	Source	Destination	Protocol	Length	Info
126	9.184055	10.103.51.159	204.79.197.200	TCP	60	59023 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
128	9.186140	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=1 Ack=1 Win=262144 Len=0
129	9.186383	10.103.51.159	204.79.197.200	TLSv1.2	543	Client Hello (SNI=www.bing.com)
132	9.188553	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=490 Ack=151 Win=261888 Len=0
133	9.188940	10.103.51.159	204.79.197.200	TLSv1.2	105	Change Cipher Spec, Encrypted Handshake Message
136	9.190291	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=541 Ack=220 Win=261888 Len=0
137	9.191125	10.103.51.159	204.79.197.200	TLSv1.2	141	Application Data
138	9.191168	10.103.51.159	204.79.197.200	TLSv1.2	92	Application Data
139	9.191267	10.103.51.159	204.79.197.200	TLSv1.2	1599	Application Data
140	9.191487	10.103.51.159	204.79.197.200	TLSv1.2	522	Application Data
141	9.191568	10.103.51.159	204.79.197.200	TLSv1.2	92	Application Data
145	9.192316	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=2717 Ack=258 Win=261632 Len=0
152	9.271056	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=2717 Ack=432 Win=261632 Len=0
326	17.696661	10.103.51.159	204.79.197.200	TLSv1.2	123	Application Data
329	17.722033	10.103.51.159	204.79.197.200	TLSv1.2	338	Application Data
334	17.790550	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=3070 Ack=1783 Win=262144 Len=0
335	17.822889	10.103.51.159	204.79.197.200	TLSv1.2	657	Application Data
339	17.840824	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=3673 Ack=3258 Win=262144 Len=0
360	17.944635	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=3673 Ack=4212 Win=261120 Len=0
361	17.946410	10.103.51.159	204.79.197.200	TLSv1.2	1212	Application Data
366	18.062718	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=4831 Ack=5236 Win=262144 Len=0
392	20.496910	10.103.51.159	204.79.197.200	TLSv1.2	186	Application Data
393	20.497124	10.103.51.159	204.79.197.200	TLSv1.2	16438	Application Data
394	20.497192	10.103.51.159	204.79.197.200	TCP	1494	59023 → 443 [ACK] Seq=21347 Ack=5236 Win=262144 Len=1440 [TCP PDU re
396	20.498303	10.103.51.159	204.79.197.200	TCP	1494	59023 → 443 [ACK] Seq=22787 Ack=5236 Win=262144 Len=1440 [TCP PDU re
398	20.498450	10.103.51.159	204.79.197.200	TLSv1.2	2655	Application Data, Application Data
415	20.580109	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=26828 Ack=5380 Win=261888 Len=0
419	20.654534	10.103.51.159	51.144.113.175	TCP	66	53495 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
423	20.726538	10.103.51.159	51.144.113.175	TCP	54	53495 → 443 [ACK] Seq=1 Ack=1 Win=262144 Len=0
424	20.728650	10.103.51.159	51.144.113.175	TLSv1.2	265	Client Hello (SNI=checkappexec.microsoft.com)
429	20.801270	10.103.51.159	51.144.113.175	TCP	54	53495 → 443 [ACK] Seq=212 Ack=1461 Win=262144 Len=0
431	20.801368	10.103.51.159	51.144.113.175	TCP	54	53495 → 443 [ACK] Seq=212 Ack=2921 Win=262144 Len=0
433	20.801500	10.103.51.159	51.144.113.175	TCP	54	53495 → 443 [ACK] Seq=212 Ack=4381 Win=262144 Len=0
435	20.801605	10.103.51.159	51.144.113.175	TCP	54	53495 → 443 [ACK] Seq=212 Ack=5841 Win=262144 Len=0

Q19. Determine the total number of packets sent to the destination IP address 204.79.197.200.

Answer:116

ip.dst==204.79.197.200						
No.	Time	Source	Destination	Protocol	Length	Info
126	9.184055	10.103.51.159	204.79.197.200	TCP	66	59023 → 443 [SYN] Seq=0 Win=65535
128	9.186140	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=1 Ack=1 Win=261888 Len=0
129	9.186383	10.103.51.159	204.79.197.200	TLSv1.2	543	Client Hello (SNI=www.bing.com)
132	9.188553	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=490 Ack=151 Win=261888 Len=0
133	9.188940	10.103.51.159	204.79.197.200	TLSv1.2	105	Change Cipher Spec, Encrypted Handshake Message
136	9.190291	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=541 Ack=220 Win=261888 Len=0
137	9.191125	10.103.51.159	204.79.197.200	TLSv1.2	141	Application Data
138	9.191168	10.103.51.159	204.79.197.200	TLSv1.2	92	Application Data
139	9.191267	10.103.51.159	204.79.197.200	TLSv1.2	1599	Application Data
140	9.191487	10.103.51.159	204.79.197.200	TLSv1.2	522	Application Data
141	9.191568	10.103.51.159	204.79.197.200	TLSv1.2	92	Application Data
145	9.192316	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=2717 Ack=258 Win=261632 Len=0
152	9.271056	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=2717 Ack=432 Win=261632 Len=0
326	17.696661	10.103.51.159	204.79.197.200	TLSv1.2	123	Application Data
329	17.722033	10.103.51.159	204.79.197.200	TLSv1.2	338	Application Data
334	17.790550	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=3070 Ack=1783 Win=262144 Len=0
335	17.822889	10.103.51.159	204.79.197.200	TLSv1.2	657	Application Data
339	17.840824	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=3673 Ack=3258 Win=262144 Len=0
360	17.944635	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=3673 Ack=4212 Win=261120 Len=0
361	17.946410	10.103.51.159	204.79.197.200	TLSv1.2	1212	Application Data
366	18.062718	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=4831 Ack=5236 Win=262144 Len=0
392	20.496910	10.103.51.159	204.79.197.200	TLSv1.2	186	Application Data
393	20.497124	10.103.51.159	204.79.197.200	TLSv1.2	16438	Application Data
394	20.497192	10.103.51.159	204.79.197.200	TCP	1494	59023 → 443 [ACK] Seq=21347 Ack=5236 Win=262144 Len=1440 [TCP PDU re
396	20.498303	10.103.51.159	204.79.197.200	TCP	1494	59023 → 443 [ACK] Seq=22787 Ack=5236 Win=262144 Len=1440 [TCP PDU re
398	20.498450	10.103.51.159	204.79.197.200	TLSv1.2	2655	Application Data, Application Data
415	20.580109	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=26828 Ack=5380 Win=261888 Len=0
624	25.622118	10.103.51.159	204.79.197.200	TLSv1.2	134	Application Data
625	25.622409	10.103.51.159	204.79.197.200	TLSv1.2	2856	Application Data
626	25.622533	10.103.51.159	204.79.197.200	TLSv1.2	92	Application Data
632	25.703259	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=29748 Ack=5
764	37.359934	10.103.51.159	204.79.197.200	TLSv1.2	134	Application Data
765	37.360116	10.103.51.159	204.79.197.200	TLSv1.2	522	Application Data
766	37.360211	10.103.51.159	204.79.197.200	TLSv1.2	92	Application Data
770	37.368690	10.103.51.159	204.79.197.200	TLSv1.2	102	Application Data

No.	Time	Source	Destination	Protocol	Length	Info
333	17.790445	204.79.197.200	10.103.51.159	TLSv1.2	92	Application Data
334	17.790550	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=36
335	17.822889	10.103.51.159	204.79.197.200	TLSv1.2	657	Application Data
336	17.824149	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=17
337	17.840682	204.79.197.200	10.103.51.159	TLSv1.2	1491	Application Data
338	17.840682	204.79.197.200	10.103.51.159	TLSv1.2	92	Application Data
339	17.840824	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=36
358	17.944532	204.79.197.200	10.103.51.159	TLSv1.2	970	Application Data
359	17.944532	204.79.197.200	10.103.51.159	TLSv1.2	92	Application Data
360	17.944635	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=36
361	17.946410	10.103.51.159	204.79.197.200	TLSv1.2	1212	Application Data
362	17.947708	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=42
364	18.062652	204.79.197.200	10.103.51.159	TLSv1.2	1040	Application Data
365	18.062652	204.79.197.200	10.103.51.159	TLSv1.2	92	Application Data
366	18.062718	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=48
392	20.496910	10.103.51.159	204.79.197.200	TLSv1.2	186	Application Data
393	20.497124	10.103.51.159	204.79.197.200	TLSv1.2	16438	Application Data
394	20.497192	10.103.51.159	204.79.197.200	TCP	1494	59023 → 443 [ACK] Seq=21
395	20.498269	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=52
396	20.498303	10.103.51.159	204.79.197.200	TCP	1494	59023 → 443 [ACK] Seq=22
397	20.498432	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=52
398	20.498450	10.103.51.159	204.79.197.200	TLSv1.2	2655	Application Data, Applic
399	20.498654	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=52
400	20.498800	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=52
401	20.498989	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=52
402	20.498989	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=52
403	20.499227	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=52
404	20.499227	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=52
405	20.499355	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=52
406	20.499501	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=52
407	20.499663	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=52
408	20.499843	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=52
409	20.499896	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=52
410	20.499963	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=52
411	20.500090	204.79.197.200	10.103.51.159	TCP	60	443 → 59023 [ACK] Seq=52

▶ Frame 333: Packet, 92 bytes on wire (736 bits), 92 bytes captured (736 bits) on interface \Device\NPF...
 ▶ Ethernet II, Src: CheckPointSo_6c:96:3f (00:1c:7f:6c:96:3f), Dst: HewlettPacka_78:cd:4a (10:60:4b:78:cd:4a)
 ▶ Destination: HewlettPacka_78:cd:4a (10:60:4b:78:cd:4a)
 ...0... = LG bit: Globally unique address (factory default)
 ...0... = IG bit: Individual address (unicast)
 ▶ Source: CheckPointSo_6c:96:3f (00:1c:7f:6c:96:3f)
 ...0... = LG bit: Globally unique address (factory default)